Power BI Basics Assignment

Question 1: What is Power BI and why is it used in businesses?

Answer

- Power BI is business intelligent tool by Microsoft.
- Helps the visualise the data and create interactive reports.
- Support the multiple data sources and real-time analytics.
- Enable to faster insights compare to statics excel reports.

Power BI why is it used in businesses

- Sales perform dashboards**:** Track sales, regions, products, and trends to monitor targets and improve revenue decisions.
- Financial revenue tracking**:** Monitor revenue, expenses, and profit trends to support budgeting and financial planning.
- Track KPI (sales, revenue, inventory).
- Spots trend and pattern in data.
- Make data-driven decision making quickly.

Question 2: Name and explain the three main components of Power BI.

Answer

Power BI three main components

Power BI desktop = build the reports.

Connect data, clean in power query, and build the reports.

Power BI services = publish & share the reports.

Services could-based platform used to publish, share, and collaborate on report created in power BI desktop.

Power BI mobile = view reports in mobile

Power BI mobile is mobile app view and interact with reports on smartphones and tables.

Question 3: Explain the Power BI workflow.

Answer

Connect = Power BI connect to different data sources to bring raw data into the system.

Transform = Data is clean, structure, and prepare analysis using power query editor.

Model = Define relationships and calculations.

Visualize (report building) = Cleaned data is converted into visual to identify the trends and patterns.

Share (insights & collaboration) = Reports publish and share with other using power BI services.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer

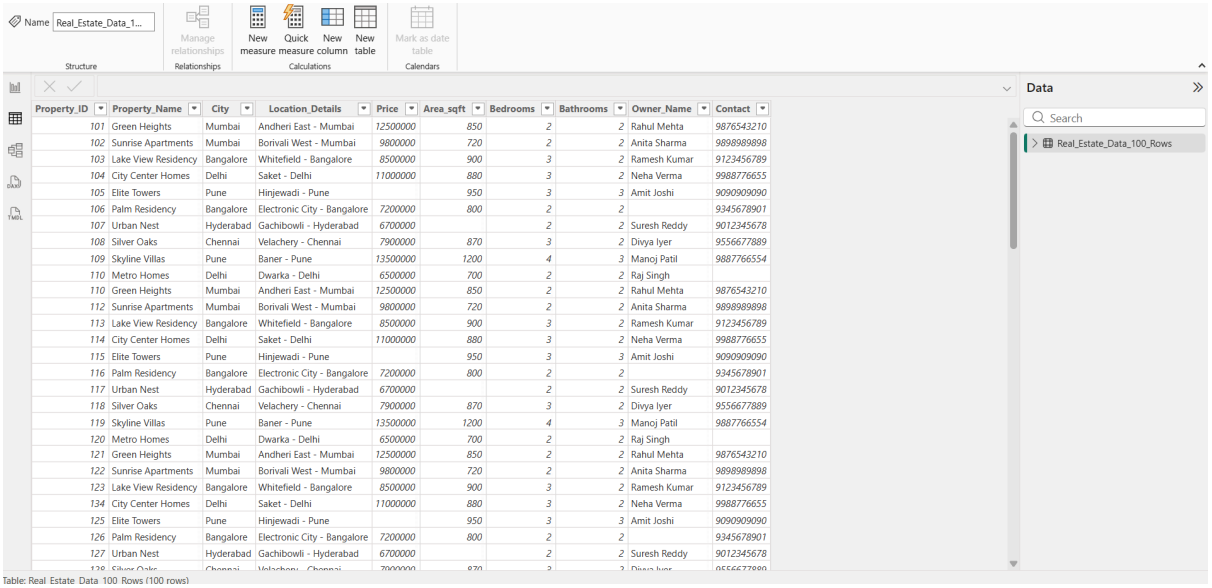
- Removing error and irrelevant data to prevent calculation failures and incorrect totals.

- Handling missing(null) value as missing values distort aggregations and visuals.

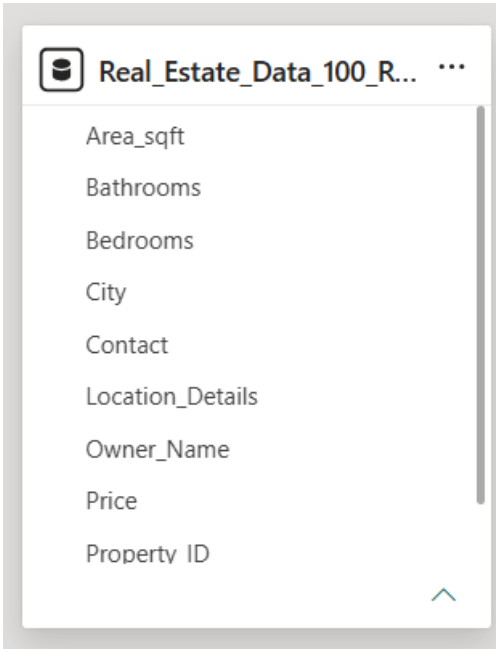
- Removing duplicates because duplicates inflate totals and misrepresent metrics.
- Standardizing data formats to ensure accurate calculations and sorting.

Question 5: Write step by step instructions to load the above dataset.

Answer



Property_ID	Property_Name	City	Location_Details	Price	Area_sqft	Bedrooms	Bathrooms	Owner_Name	Contact
101	Green Heights	Mumbai	Andheri East - Mumbai	12500000	850	2	2	Rahul Mehta	9876543210
102	Sunrise Apartments	Mumbai	Borivali West - Mumbai	9800000	720	2	2	Anita Sharma	9898989898
103	Lake View Residency	Bangalore	Whitefield - Bangalore	8500000	900	3	2	Ramesh Kumar	9123456789
104	City Center Homes	Delhi	Saket - Delhi	11000000	880	3	2	Neha Verma	9988776655
105	Elite Towers	Pune	Hinjewadi - Pune		950	3	3	Amit Joshi	9090909090
106	Palm Residency	Bangalore	Electronic City - Bangalore	7200000	800	2	2		9345678901
107	Urban Nest	Hyderabad	Gachibowli - Hyderabad	6700000		2	2	Suresh Reddy	9012345678
108	Silver Oaks	Chennai	Velachery - Chennai	7900000	870	3	2	Divya Iyer	9556677889
109	Skyline Villas	Pune	Baner - Pune	13500000	1200	4	3	Manoj Patil	9887766554
110	Metro Homes	Delhi	Dwarka - Delhi	6500000	700	2	2	Raj Singh	
110	Green Heights	Mumbai	Andheri East - Mumbai	12500000	850	2	2	Rahul Mehta	9876543210
112	Sunrise Apartments	Mumbai	Borivali West - Mumbai	9800000	720	2	2	Anita Sharma	9898989898
113	Lake View Residency	Bangalore	Whitefield - Bangalore	8500000	900	3	2	Ramesh Kumar	9123456789
114	City Center Homes	Delhi	Saket - Delhi	11000000	880	3	2	Neha Verma	9988776655
115	Elite Towers	Pune	Hinjewadi - Pune		950	3	3	Amit Joshi	9090909090
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Question 6: Define Data View, Report View, and Model View. Explain the purpose of each view.

Answer

Report view = create and design report, design charts, tables, and dashboards.

Data view = view the imported data, Shows the raw table data after transformations.

Model view = manage the relationship between the table (like Property_ID linking multiple datasets).

Question 7: Discuss the different data sources that Power BI supports.

Answer

- **Files**: Excel, CSV, XML, JSON
- **Databases**: SQL Server, Oracle, MySQL, PostgreSQL
- **Cloud Services**: Azure, Google Analytics, Salesforce
- **Web APIs**: REST APIs, SharePoint, etc.

Question 8: Split Owner Name to create two new columns as First Name and Last Name.

Answer



	Value.1	Value.2
	Rahul	Mehta
	Anita	Sharma
	Ramesh	Kumar
	Neha	Verma
	Amit	Joshi
		null
	Suresh	Reddy
	Divya	Iyer
	Manoj	Patil
	Raj	Singh
	Rahul	Mehta
	Anita	Sharma
	Ramesh	Kumar
	Neha	Verma
	Amit	Joshi
		null
	Suresh	Reddy
	Divya	Iyer
	Manoj	Patil
	Raj	Singh
	Rahul	Mehta
	Anita	Sharma
	Ramesh	Kumar
	Neha	Verma
	Amit	Joshi
		null
	Suresh	Reddy